Research Safety Summary - View

RSS No

4198.16

Status

Authorized

Minor Revisions

None.

Last Modified by

Cooper, Stan (27714) on 8/12/2024 10:10:55 AM

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Title

Research Safety Summary for the Spallation Neutron Source Extended-Q Small Angle Neutron Scattering (EQ-SANS) Diffractometer Beamline 6.

Approvals/Authorizations

Date/Time	Name	Role
6/5/2024 9:20:07 AM	Gao, Carrie Y (900864)	Approver
6/5/2024 10:33:59 AM	Do, Changwoo (977443)	Approver
6/5/2024 5:51:58 PM	Nagy, Gergely (3072000)	Approver
6/6/2024 7:28:29 AM	Hamby, Kevin (937079)	Authorizer

Description Text

This summary tabulates hazards encountered when working on the Extended-Q Small Angle Neutron Scattering (EQ-SANS) diffractometer on beam line 6 of the Spallation Neutron Source Target Building (Building #8700). The EQ-SANS Diffractometer (see Fig. 1) is designed to study non-crystalline, nano-sized materials in solid, liquid, or gas forms. All individual measurements will be subject to an Experimental Safety Review - which examines the safety and hazards, including combustibility associated with individual sample materials. This review will be done in accordance with the requirements of the Spallation Neutron Source, Final Safety Analysis Document - Neutron Facilities and applicable ORNL SBMS procedures. This RSS also covers the work spaces associated with the work benches outside the instrument hutch and at the sample exchanger.

Note: This RSS does not authorize work or establish the hazards and controls for work in the attached laboratory space. Check with the lab's LSM, Koldys, Alex (615-823-0310), for requirements for work in that space. Any activity(ies) outside the scope of Research & Development (R&D), all preventative maintenance, and all corrective maintenance shall be performed in accordance with the Chestnut Ridge Operations, Maintenance, and Services Work Control Procedure.

Description File

None.

Division

X186: Neutron Scattering Division

Start Date

12/19/2005

End Date

None.

Authorization Period

One Year

Account

None.

General Notes

Sample/task specific hazard analysis may be prepared to cover task/sample specific hazards or addition detail that is not contained in the RSS.

Discharge of incidental wastewaters produced during operation of the EQ-SANS instrument are covered by the variance described in the attached PDF document.

General Attachments

Review of SS Variance UTB-127 Permanent SNS Incidental Discharges Feb 2024.pdf

Point of Contact

Fagan, Lisa (742561)

Authorizer

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Principal Investigator

Do, Changwoo (977443);

PI Delegates

Gao, Carrie Y (900864); Heller, William T (901822); Nagy, Gergely (3072000);

Approvers

Do, Changwoo (977443) - *Principal Investigator*; Gao, Carrie Y (900864) - *Lab Space Manager*; Nagy, Gergely (3072000) - *PI Delegate*;

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Required Reviewers

Elam, Christi L (3052977) - *NRPD*; Fagan, Lisa (742561) - *QHSP*; Heller, William T (901822) - *Group Leader*; Hodges, Ryan (3015252) - *Environmental Compliance Representative*; Parrott Jr, Danny (923220) - *DESO*;

Optional Reviewers

Davison, Brian (27002) - *ORNL Institutional Biosafety Committee (Chair)*; Duff, Jeffrey (34088) - *IHC Group Leader*; Everett, Michelle (961144) - *SA Group Leader*; Gao, Carrie Y (900864) - *Lab Space Manager*; Greeley, Leigh (29033) - *Human Subjects Research Coordinator*; Hamby, Kevin (937079) - *Section Head*; Hostetler, Aaron (3120594); Martin, Joel (51443) - *Security Contact (Alternate)*; Tache, Julie (945839) - *Security Contact (Alternate)*; Urban, Volker S (902316) - *Section Head*; Wentzel, Taylor (3100103) - *Pressure Systems SME*; Williams, Justin (3088930) - *Fire Protection Engineer*;

Lab Space Managers

Gao, Carrie Y (900864)

Locations

Building 8700, Room TA-2061

Nepa Documentation

Approval for this project and/or activity is granted by the Record of Decision for the Construction and Operation of the Spallation Neutron Source, issued June 30, 1999. The respective ROD is based on the analysis contained in the "Final Environmental Impact Statement for the Construction and Operation of the Spallation Neutron Source" (SNS, FEIS, DOE/EIS-0247, April 23, 1999)

Hazards

First

This operation requires specific programmatic controls to protect information (Classified Subject Areas, Export Control, Intellectual Property, or CRADAs).

Requirements:

Agreements for Commercializing Technology (ACT) Projects
Cooperative Research and Development Agreements (CRADAs)
Export Control Compliance
Information Protection
Proprietary Information

Hazard Notes:

R&D occasionally has the potential to develop new materials or processes that are potentially patentable.

Control Notes:

Participants should be aware of opportunities to develop intellectual property and should consult with the group leader if questions arise regarding intellectual property.

Locations:

All Locations

Attachments:

None.

1.1

This work involves the use of a particle accelerator as a user.

Hazard Notes:

While the Spallation Neutron Source is an accelerator facility, the operation of the EQ-SANS (BL-6) does not involve direct use of the accelerator. The only direct interaction the instrument has with the accelerator is via the Instrument Personel Protection System designed to protect operators of the instrument from prompt radiation. The IPPS can initiate a sequence of events that halts the operation of the accelerator.

Control Notes:

Does not directly apply to the operation of EQ-SANS (BL-6)

Locations:

All Locations

Attachments:

None.

1.2

This work involves responsibility for oversight, operation, or management of a particle accelerator.

Requirements:

Accelerator Safety

Hazard Notes:

EQ-SANS (BL-6) includes an incident flight path that directs a neutron beam generated in the Target Facility to a sample. Direct exposure to this beam at the sample location could result in a dose level over 200 mrem/hr to an extremity (hand) or possibly a whole body dose for a person in the final flight path.

Control Notes:

The radiation area generated by the incident flight path is only a hazard when the shutter for the instrument is open. The open shutter condition is controlled by the IPPS, a credited engineering control implemented on all instruments at the SNS. The IPPS sufficiently controls access to the room where the neutron flight path exists, and has the capability of closing the shutter and/or disabling the accelerator in the event that these controls are breached. Radiological shielding is controlled by configuration locks, cables and tags as deemed appropriate by the SNS RSO, Instrument Staff, and NSSD ESH/Ops Manager.

Locations:

All Locations

Attachments:

None.

2.1

This operation involves the generation, handling, processing, use, or storage of radioactive materials (including sealed sources and waste).

Requirements:

Facility Hazard Categorization
Facility Use Agreements
HEPA Filters Used in Fixed Nuclear/Radiological Applications
Radiological Area Controls

Radiological Design Requirements
Radiological Dosimetry
Radiological Work
Safety and Security Regulatory (SSR) Program
Sealed Radioactive Source Control

Hazard Notes:

Samples and sample environment equipment will be exposed to a flux of neutrons during normal operation of the instrument. In addition, some equipment associated with the instrument (neutron choppers) will require periodic maintenance and will be activated. Some free liquid samples may consist of accountable quantities of deuterated water or other solvents.

Control Notes:

Samples are managed in accordance with the procedure "NScD Sample Management at SNS & HFIR" at https://recordsmgmt.ornl.gov/DMS_ORNL/DocumentUpload/20000/20050-3-0.pdf.

Depending upon the conditions for post irradiation sample handling and manipulation, staff will be complete for one of the following curriculum requirements prior to engaging in activities:

- ***Radiological Worker I for ORNL** and **RadEye G User (Staff)***
- ***Radiological Worker II for ORNL** and **RadEye G User (Staff)***
- ***Rad Worker for Neutron Scattering Research***

All potential wastes generated by this operation will be identified and evaluated by the Principal Investigator and the SNS Environmental Protection Officer to ensure that newly generated wastes have disposal paths, and that accurate forecasts are developed and maintained to support disposition of those operational wastes.

Locations:

All Locations

Attachments:

None.

3.1

This operation involves any of the following NIR sources:

- permanently installed microwave/radio-frequency gear (including induction heaters and communications transmitters) capable of radiating over 1 W into an open area at frequencies between 3 kHz and 300 GHz or of emitting over 100 W if the output is normally completely enclosed by coaxial cables, waveguides, or dummy or real loads
- portable walkie-talkie communications sets capable of radiating over 7 W at frequencies between 100 kHz and 450 MHz, and over 7 (450/f) W at frequencies between 450 MHz and 1.0 GHz (f in MHz)
- static magnetic fields which may exceed 0.5mT (5 gauss)
- sub-radiofrequency (30 kHz and below) magnetic fields exceeding 0.1 mT (1 gauss) or electromagnetic fields exceeding 1842 V/m.
- any equipment that would expose personnel to high levels of Visible Light (400 nm – 770 nm) at >1 candela/cm2; to high levels (>30 J/m2) of Near Infrared (770nm-3000nm) at >10mW/cm2; and/or high levels of Ultraviolet (UV) radiation (180nm-400nm)
- any infrared heat lamp or any near-infrared source where a strong visual stimulus is absent (luminance of less than E-06 candela/cm2)

Requirements:

Conduct Exposure Assessments
Work with Nonionizing Radiation

Hazard Notes:

Sample environments used for neutron scattering experiments may involve one or more of the aforementioned NIR sources. Such hazards are identified during the proposal safety review process.

Control Notes:

Potential NIR sources may be used inside the sample chamber with steel enclosure. All experiments are reviewed for safety and appropriate hazard controls are identified and implemented. All hazard controls identified and developed during the experimental safety review must be followed by beamline personnel and visiting scientists.

Locations:

All Locations

Attachments:

None.

5.2

This operation involves work on equipment that requires Lock/Tag/Verify control procedures (equipment has the potential to release hazardous energy, e.g. electrical, mechanical, pressure, steam).

Requirements:

Electrical Safety
Lock/Tag/Verify

Hazard Notes:

LO/TO operation will not be performed by beamline personnel.

Control Notes:

LO/TO will be performed by trained and authorized personnel only. The instrument staff will be required to understand when LO/TO is required.

Locations:

All Locations

Attachments:

None.

5.4

This operation involves exposure to one or more of the following electrical hazard sources:

- Direct current (DC) sources;
- Batteries, other than household batteries;
- Capacitors;
- Unguarded inductive systems that pose a hazard.

Requirements:

Electrical Safety

Hazard Notes:

Sample environments used for neutron scattering experiments may involve one or more of the aforementioned electrical hazard sources. Such hazards are identified during the proposal safety review process.

Control Notes:

All experiments are reviewed for safety and appropriate hazard controls are identified and implemented. All hazard controls identified and developed during the experimental safety review must be followed by beamline personnel and visiting scientists.

Locations:

All Locations

Attachments:

None.

6.1

This operation involves exposure to, or use of, recombinant DNA.

Requirements:

Biohazards

Hazard Notes:

Recombinant DNA samples may be measured on the EQ-SANS instrument. Most material manipulations will occur in the adjacent laboratories and are covered by the controls specified in the apporparite RSS for the space employed.

Sample material is identified during the experimental safety review process and appropriate approvals obtained. All identified standard biosafety practices required after review will be employed.

This RSS overlaps with RSS for the adjacent laboratories but manipulation of samples occurs in the labs or consists of sample disposal in an RMA area outside the lab.

Control Notes:

All experimental sample handling will be compliant with all SBMS controls put forward in the BIOHAZARDS subject area.

There is an experiment safety review that will perform the screening of the materials proposed. This review will determine whether or not further review by the ORNL IBC is required for this particular experiment. Additional requirements may be necessary. Standard Biosafety level 1 Standard will be used unless higher levels are required upon review.

Specific Requirements:

* Persons wash their hands after they handle viable materials and/or animals, after removing gloves, and before leaving the laboratory.

* Mouth pipetting is prohibited; mechanical devices are used.

* Careful management of needles and other sharps are of primary importance. Needles must not be bent, sheared, broken, recapped, removed from disposable syringes, or otherwise manipulated by hand before disposal. Used disposable needles and syringes must be placed in puncture-resistant containers. Place non disposable sharps a hard-walled container when not in use or when being transported. Broken glassware must not be handled directly without cut-resistant gloves. Remove using a brush and dustpan, tongs, or forceps.

* All procedures are performed carefully to minimize the creation of the splashes or aerosols.

* All sample materials and other regulated wastes are decontaminated before disposal per SBMS procedures by an approved decontamination method, such as autoclaving or bleach (>10%) killing (see the Disposing of Biological Agents section for other decontamination and disposal options). All other samples will be kept contained and returned to the originator.

* Staff will contact their supervisor and report to the Occupational Health Services Center if they believe they have been exposed to any biohazard.

* Blunt tip needles will be preferably used.

Work with similar materials in laboratories not currently covered by the registration with the Biosafety Review Committee must be approved, prior to start-up.

Whenever possible, samples of this nature will be delivered to the beamline loaded into sealed sample cells for measurements and kept sealed. A spill control plan, which will include a detailed procedure, lists of required contacts in the event of a spill and the specific contents and location of a spill kit (such as gloves, a squirt bottle with 70% ethanol or 10% bleach, and absorbent wipes, etc. as appropriate for the specific hazards present), will be developed to cover spills and any opening of the sample. Further controls include assignment of appropriately designated, suitable lab space. Any samples identified to be BSL2 will be considered for additional controls, double containment and additional spill response procedures.

A plan for the final disposition of the samples, whether they are to be returned unopened to the user's institution or disposed of at ORNL, must be agreed upon prior to final approval of the neutron scattering experiments.

Locations:

All Locations

Attachments:

None.

6.2

This operation involves exposure to, or handling of, human tissues or body fluids, or human cells in culture.

Requirements:

Biohazards
Human Subjects Research

Hazard Notes:

Experimental Samples for use on the Neutron Scattering Instruments may be human tissues, body fluids or cells.

Sample materials identified to fall into this category during the experimental safety review process will be sent to the ORNL IBC for additional review.

All appropriate BSL sample handling practices and controls recommended after review will be employed.

Control Notes:

This RSS overlaps with other RSS's that cover work performed in the sample preparation laboratories. All experimental sample handling will be compliant with SBMS controls put forward in the BIOHAZARDS subject area.

There is an experiment safety review process that will perform the initial screening of the materials proposed. Additional review by the ORNL IBC is required for experiments utilizing these materials and additional sample handling requirements may be necessary.

The IBC will determine the appropriate BSL level prior to intiation of work.

The ORNL Human Subjects Research Coordinator will need to be notified of any projects involving human tissue/cells/fluids regardless of the source (commercial or otherwise), prior to commencement of work. The Coordinator will work with the IRB offices to determine what level of review will be necessary depending on the type of sample.

Whenever possible, samples of this nature will be delivered to the beamline loaded into sealed sample cells for measurements and kept sealed. A spill control plan will be developed to cover spills and any opening of the sample will require further controls including appropriately designated lab space. Any samples identified to be BSL2 will be considered for additional controls, double containment and spill response. Samples of human origin that require manipulation or open container work that meets OSHA's definition of potential for exposure found in the Bloodborne Pathogen standard may be subject to full requirements of the standard including but not limited to a written Exposure Control Plan.

A plan for the final disposition of the samples, whether they are to be returned unopened to the user's institution or disposed of at ORNL, must be agreed upon prior to final approval of the neutron scattering experiments.

Specific Requirements: For lab work see controls in question 6.1

Locations:

All Locations

Attachments:

None.

6.3

This operation involves exposure to, or handling of, Risk Group 1 microorganisms, dead or alive.

Requirements:

Biohazards

Hazard Notes:

Experimental samples measured using the EQ-SANS instrument may consist of Risk group 1 organisms, dead or alive.

Sample materials identified to fall into this category during the experimental safety review process will be sent to the ORNL IBC for additional review.

All appropriate BSL sample handling practices and controls recommended after review will be employed.

Control Notes:

This RSS overlaps with other RSS's that cover work performed in the sample preparation laboratories. All experimental sample handling will be compliant with SBMS controls put forward in the BIOHAZARDS subject area.

There is an experiment safety review process that will perform the initial screening of the materials proposed. Additional review by the ORNL IBC is required for experiments utilizing these materials and additional sample handling requirements may be necessary.

The IBC will determine the appropriate BSL level prior to intiation of work.

Whenever possible, samples of this nature will be delivered to the beamline loaded into sealed sample cells for measurements and kept sealed. A spill control plan will be developed to cover spills and any opening of the sample will require further controls including appropriately designated lab space. Any samples identified to be BSL2 will be considered for additional controls, double containment and spill response.

A plan for the final disposition of the samples, whether they are to be returned unopened to the user's institution or disposed of at ORNL, must be agreed upon prior to final approval of the neutron scattering experiments.

Specific Requirements: For lab work see controls in question 6.1

Locations:

All Locations

Attachments:

None.

6.5

This operation involves biological select agents or toxins, or biological agents, dead or alive.

Requirements:

Biohazards

Hazard Notes:

Select agents and toxins may be samples measured using the EQ-SANS instrument.

Sample material is identified during the experimental safety review process and appropriate approvals obtained. All identified biosafety practices required after review will be employed.

This RSS overlaps with RSS for the adjacent laboratories but manipulation of samples occurs in the labs or consists of sample disposal in an RMA area outside the lab.

Control Notes:

All experimental sample handling will be compliant with all SBMS controls put forward in the BIOHAZARDS subject area.

There is an experiment safety review that will perform the screening of the materials proposed. This review will determine whether or not further review by the ORNL IBC is required for this particular experiment. Additional procedures and controls may be necessary.

Work with materials in laboratories not currently covered by the registration with the Biosafety Review Committee must be approved, prior to start-up.

Whenever possible, samples of this nature will be delivered to the beamline loaded into sealed sample cells for measurements and kept sealed. A spill control plan will be developed to cover spills and any opening of the sample will require further controls including appropriately designated lab space. Certain listed toxins can be handled at de minimus levels below the regualtory limit with appropriate safety controls and review. Additional controls, double containment and additional spill response measured will be considered and employed, as appropriate.

A plan for the final disposition of the samples, whether they are to be returned unopened to the user's institution or disposed of at ORNL, must be agreed upon prior to final approval of the neutron scattering experiments.

Note that no select agent or toxin can be used without Center for Disease Control (CDC) registration, which will require considerable time and effort that must be accounted for in the experiment planning and scheduling. IBC reviewed and approved select toxins may be used only at a de minimus level without official CDC registration.

Locations:

All Locations

Attachments:

None.

6.6

This operation involves research with plant pests, plant pathogens, or invasive plants.

Requirements:

Biohazards

Hazard Notes:

Some user samples may contain organisms , plants, tissues or soils that contain plant pathogens at BL-1p or BL-2p levels.

Control Notes:

Work with USDA-regulated organisms may be subject to additional controls specified under permit conditions. **Participation in this RSS alone does not authorize work under permits.**

Locations:

All Locations

Attachments:

None.

7.2

This operation involves work conducted under the OSHA Hazard Communication Program (HAZCOM).

Requirements:

Chemical Safety

Hazard Notes:

Chemical hazards from sample materials.

Control Notes:

1.) Common experimental activities will be covered by the Experimental Saftey Review Process.

2.) Prior to use on the instrument, all samples/experiments will be subject to an experimental review through the Proposal Management System at the SNS.

3.) Use of samples at the instrument will also be considered as part of a job hazards analysis. Consideration of the relevant MSDS is part of this process.

4.) All potential wastes generated by this operation will be identified and evaluated by the Principal Investigator and the SNS Environmental Protection Officer to ensure that newly generated wastes have disposal paths and that accurate forecasts are developed and maintained to support disposition of those operational wastes.

5.)ORNL HazCom training will be required. All materials will be labeled in accordance with the HazCom standard.

6.) Limited quantities of alcohols and ketones (<500 ml each) are used periodically to clean materials. Review of the quantity (<500 ml) and frequency (spread out over a month) indicates the exposure risk is acceptable and can be considered "low."

7) Representative monitoring of respirable material has shown levels equal to or less than background levels when handling small amounts (e.g. ~5 grams) of powder samples. This information coupled with handling times often in minutes (e.g. 1-4) and the use of gloves to prevent dermal exposure justifies the exposure risk as acceptable or “low risk.”.

8) Liquid samples are typically analyzed at this beamline and are composed primarily of low toxicity , low vapor pressure materials with volumes usually in the < 2 ml range. Sample preparation activities do not involve activities such as sonication, shaking, and mixing and therefore do not increase the likelihood of generating aerosols. This information coupled with handling times often in minutes (e.g. 1-4) and the use of gloves to prevent dermal exposure justifies the exposure risk as acceptable or “low risk.”

Locations:

All Locations

Attachments:

None.

7.3

This operation involves any chemicals or wastes that present health hazards (carcinogens,reproductive hazards, toxics, or toxics with potential for skin absorption).

Requirements:

Chemical Safety
Conduct Exposure Assessments
Personal Protective Equipment
Radiological Monitoring of Individuals and Areas

Hazard Notes:

Chemicals that could present health hazards such as carcinogens, mutagens, toxins or cytotoxic drugs could be used in research experiments.

Control Notes:

Precautions must be taken to eliminate exposure to these materials. Gloves, safety glasses, hoods, etc. are appropriate for controlling exposure to these materials. Where possible, alternatives should be used.

Steps must be taken to eliminate skin contact and other types of exposures to carcinogens. Frequency, duration, concentration and experimental conditions should be evaluated along with the specific chemical. Use of hoods, appropriate PPE specific to the chemical, dilution, and minimizing handling time are minimal controls for using carcinogens.

NOTE: See List of Proposition 65 Chemicals, also known as the "California List" of carcinogens. This list is a very conservative (note that toluene is listed) list of chemical identified by legislation in California. ORNL identifies carcinogens in HMMS when their concentration > 0.1% and they are also listed by International Agency for Research on Cancer (IARC), National Toxicology Program (NTP)(See pdf attachment to this question for list), American Conference of Industrial Hygienists (ACGIH), and OSHA. If you have any questions about these lists, contact the Lab Space Manager, the Principal Investigator, or the Divisions ES&H representative.

If these materials are used, and the controls listed above are not adequate, then a task based job hazard analysis shall be completed.

If working with cadmiums to make custom-masks to block neutrons beams, cadmium worker training (CRS 00006778 training) should be completed prior to the job.

Any wastes generated are handled in accordance with ORNL SBMS requirements by RCRA/SAA trained custodian.

Locations:

All Locations

Attachments:

tr371.pdf

7.5

This operation involves chemicals or wastes that are pyrophoric.

Requirements:

Chemical Safety
Personal Protective Equipment

Hazard Notes:

Experimental samples of lithium, sodium and other pyrophoric metals may be present. These materials will be identified through the experimental safety review process.

Control Notes:

- Samples are small (2-10mls average) and stored under oil or vacuum. Proper extinguishers are available in the area where these materials are handled. Any special storage and handling requirements shall be included in the required safety review of experiments.
- Use of these materials may also require a review by fire protection engineering before approved for use.
- Any wastes generated are handled in accordance with ORNL SBMS requirements by RCRA/SAA trained custodian.
- Samples of this nature will be stored in the glove box in the storage room in the basement after the completion of experiment. Contact Instrument team for guidance.

Locations:

All Locations

Attachments:

None.

7.6

This operation involves chemicals or wastes that are time, condition, or shock sensitive (e.g. peroxide-forming chemicals, picric acid, etc.).

Requirements:

Chemical Safety
Personal Protective Equipment

Hazard Notes:

Small quantities of shock sensitive chemicals are used as samples.

Control Notes:

- Small lab quantities are used and stored properly.
- Any wastes generated are handled in accordance with ORNL SBMS and /or other requirements.
- Proper fire extinguishers are provided in the area.
- Samples of this nature will be stored in the glove box in the storage room in the basement after the completion of experiment. Contact Instrument team for guidance.

Locations:

All Locations

Attachments:

None.

7.7

This operation involves any chemicals or waste that are flammable liquids, flammable gases, flammable solids, combustible liquids, combustible metals, or combustible dusts.

Requirements:

Chemical Safety
Compressed Gas Cylinders and Related Systems
Fire Protection, Prevention & Control
Personal Protective Equipment

Hazard Notes:

See notes under section 7.3

Control Notes:

See controls under section 7.3.

Flammable materials other than routine sample volumes should not be stored at the beam line. Storage of bulk materials is available in the CLO or the South Side User Lab.

Locations:

All Locations

Attachments:

None.

7.8

This operation involves chemicals or wastes that are caustic or corrosive (e.g. acids or bases).

Requirements:

Chemical Safety
Conduct Exposure Assessments
Personal Protective Equipment

Hazard Notes:

See notes under section 7.3

Control Notes:

See notes under section 7.3

Locations:

All Locations

Attachments:

None.

7.14

This research involves the use of engineered nanoparticles.

Requirements:

Chemical Safety
Conduct Exposure Assessments
Respiratory Protection
Unbound Engineered Nanoparticles

Hazard Notes:

Small quantities of engineered nanoparticles may be the subject of scattering experiments. These are in most cases either in solution or imbedded particles.

Control Notes:

All operations involving nanoparticles will be performed in rigid compliance with the specifications in the Using Engineered Nanomaterials in Chemical Safety subject area in SBMS https://sbms.ornl.gov/sbms/SBMSearch/SubjArea/Nano/pro2.cfm

Individual experiments shall be reviewed through the NSSD proposal system. Nano materials in free-state require a job specific safety plan or hazard analysis before the work is scheduled.

Locations:

All Locations

Attachments:

None.

8.2

This operation generates hazardous waste.

Requirements:

Environmental Management
Manage Waste and Excess Materials

Hazard Notes:

Spent solvents or toxic metals such as cadmium may be generated

Control Notes:

Proper identification, characterization and storage of waste is critical to the compliant operation of ORNL. Always contact the division ECR or WSR for waste determination assistance when waste generating activity involving hazardous constituents is planned. Notification should take place before the date waste is generated.

Hazardous wastes must be collected and stored in an RCRA Satellite Accumulation Area (SAA) at or near the point of generation. Wastes must be labeled with the following:

- The words "Hazardous Waste" plus an accurate description of the contents
- A hazard indication. Examples are GHS pictograms or indicator words such as "Ignitable" or "Toxic" or "Corrosive" etc.
- Contact name and phone number optional but helpful

SAAs must be registered with the Environmental Protection Services Division (EPSD), and the SAA sign must be posted. See the division ECR for assistance managing SAAs.

No more than 55 gallons of hazardous waste OR one quart of liquid acute hazardous waste OR one Kg of solid acute hazardous waste may be stored on a SAA at one time. Contact the division ECR if these limits will be exceeded.

Every waste container or addition to a container must be entered in the SAA log.

Waste storage containers shall be compatible with the waste and closed at all times.

SAA custodians and alternates must have current RCRA Satellite Area at ORNL training. Contact the Division Training Officer for information.

Wastes in an SAA must be segregated for compatibility. Anyone adding waste to an SAA must be approved by the SAA custodian.

No treatment of hazardous waste is allowed without approval from ORNL's Office of Environmental Protection.

Locations:

All Locations

Attachments:

None.

8.3

This operation generates radioactive waste (including mixed hazardous, solid low level, liquid low level, transuranic, or other radioactive waste with no identified path for disposal).

Requirements:

Comply with Clean Water Act
Environmental Management
Manage Waste and Excess Materials

Hazard Notes:

Activated materials may be/ are produced by neutron irradiation of samples and laboratory equipment. Some of these materials may be transferred to the laboratories.

Radioactive material is not permitted in all lab spaces. Check Posting and LSM to verify.

Control Notes:

see 8.2

All applicable ORNL/SBMS requirements and procedures are followed for handling and disposal of activated radioactive material, in addition to those for hazardous waste. Participants should consult with the Laboratory Space Manager if with questions or concerns.

Locations:

All Locations

Attachments:

None.

8.4

This operation generates medical or biohazardous waste.

Requirements:

Biohazards
Environmental Management
Manage Waste and Excess Materials

Hazard Notes:

The potential for producing biohazardous waste exists as a result of experimental activities using the EQSANS.

The experimental safety review will identify if the potential is there and set appropriate controls.

Control Notes:

The procedure NScD Sample Management at SNS & HFIR requires that the user of the neutron scattering facilities specify whether ORNL is to dispose of waste materials resulting from their experiment, or if the samples are to be returned to the originating lab for proper disposal. The sample management procedure is available for review at

Sample Management Procedure

If biohazardous sample wastes are to be disposed of at ORNL, all waste will be sterilized (either with bleach, ethanol, or autoclave) using a disinfection/deactivation method that is recognized as effective for the specific biological material and contained as recommended. Use and maintenance of the autoclave shall be covered by the manufacturer’s documentation and manuals. All participants using the autoclave shall be aware of the hazards of using an autoclave and only operate it in accordance with the manufacturer's instructions.

WSR will be consulted if questions arise.

Locations:

All Locations

Attachments:

None.

8.6

This operation generates special waste.

Requirements:

Environmental Management
Manage Waste and Excess Materials

Hazard Notes:

Aerosol cans may be utilized.

Control Notes:

Aerosol cans will be disposed of via guidance from the waste representative.

Locations:

All Locations

Attachments:

None.

8.7

This operation generates recyclable material (used oil, scrap metal, universal waste, photographic waste).

Requirements:

Environmental Management
Manage Waste and Excess Materials
Prevent Pollution

Hazard Notes:

Recyclable materials potentially generated include used oil, lamps/bulbs, batteries, and scrap metal.

Control Notes:

- Used oil shall have the words "used oil" written on the container and logged into a used oil collection point. If oil is to be re-used in processes in the lab, label the container indicating this so as not to be confused with oil ready for recycle. Silicone based oils are collected separately from hydrocarbon based oils
- Metals that have never been in a radiological material management area can be placed in the metal recycle
- Alkaline batteries. light bulbs and printer cartridges should be taken to the appropriate recycle locations.
- Contact NScD Environmental Protection Representative/Waste Representative for assistance if needed.

Locations:

All Locations

Attachments:

None.

9.1

This operation involves activities requiring the use of mechanically-assisted lifting (e.g. pallet jacks, lift tables, fork lifts), rigging, or hoisting.

Requirements:

Material Handling

Hazard Notes:

Some optics or sample environment equipment (e.g. collimators, exchangeable sections of guide, variable apertures) must be transported to the sample area and lowered into position using large, overhead crane at the SANS beamline.

Control Notes:

1) SNS and/or ORNL institutional hoisting and rigging training is required.

2) The crane is not be be operated by users or other untrained staff. Operation of the crane will be controlled by disabling the crane control box or through lockout of the crane disconnect switch.

Locations:

All Locations

Attachments:

None.

9.2

This operation involves exposure to moving or rotating parts, such as motors, shafts, pulleys, belts, or any other potential mechanical energy.

Requirements:

Occupational Hazard Controls

Hazard Notes:

The various sample environments used at the beam line have moving parts.

Movement of large parts of the instrument, such as the beamstop and detector tank, are only performed by qualified personnel.

Control Notes:

Follow all beam line procedures for sample changes.

The sample environments with moving parts are generally not physically large (all presently used ones weigh less than 100 pounds) and the movements are relatively slow.

When maintenance requires movement of large parts of the instrument, such as the beamstop and detector tank, all non-essential personnel should vacate the immediate area unless authorized to be present to observe the work by the qualified personnel performing the work.

Locations:

All Locations

Attachments:

None.

9.3

This operation involves handling sharp objects that could result in cuts, abrasions, or punctures to the hands and/or arms.

Requirements:

Personal Protective Equipment

Hazard Notes:

Cutting tools, razor blades, and syringe needles are used in the laboratories.

Control Notes:

When using cutting tools with open blades, keep the cutting blade moving away from fingers, hands, and the body. When not in use, conceal the blade (cut resistant container or blade cover, etc.).

- Cut resistant gloves should be considered for all activities where cutting tools of any type are used
- Razor blades, scalpels and other fixed blade or manually retractable blade knives must be disposed of in an approved sharps container
- Properly handle needle-attached syringes to prevent poking hands, eyes and others. Never apply force to the needle towards oneself or others.
- Recapping needles using is not recommended.
- Do not overfill sharps containers.

Locations:

All Locations

Attachments:

None.

11.1

This operation involves the use of a High-Energy Pressure System(s).

Requirements:

Compressed Gas Cylinders and Related Systems
Engineering Design
Pressure System Safety

Hazard Notes:

Sample environments may involve high-energy pressure systems.

Control Notes:

All experiments are reviewed for safety and appropriate hazard controls are identified and implemented. All hazard controls identified and developed during the experimental safety review must be followed by beamline personnel and visiting scientists.

Locations:

All Locations

Attachments:

None.

11.2

This operation involves the use of a Moderate-Energy Pressure System(s).

Requirements:

Compressed Gas Cylinders and Related Systems
Engineering Design
Pressure System Safety

Hazard Notes:

Sample environments may involve moderate-energy pressure systems.

Control Notes:

Please refer to the control notes in section 11.1

Locations:

All Locations

Attachments:

None.

11.3

This operation involves other processes that might create a pressure hazard not categorized or considered as part of a High or Moderate-Energy system(s).

Requirements:

Compressed Gas Cylinders and Related Systems
Engineering Design
Pressure System Safety

Hazard Notes:

Sample environments may involve a pressure hazard.

Control Notes:

Please refer to the control notes in section 11.1.

Locations:

All Locations

Attachments:

None.

11.5

This operation involves hazards associated with the physical handling or storage of compressed gas cylinders.

Requirements:

Compressed Gas Cylinders and Related Systems

Hazard Notes:

Compressed air may be used in sample handling. Users may include the use of compressed gases in experiments.

Control Notes:

1) ORNL training in the use of compressed gas cylinders will be required for all SANS staff members.

2) Use of compressed gases by users will be identified by the experimental safety summary.

Locations:

All Locations

Attachments:

None.

12.2

This operation involves cryogenics or other cold thermal hazards that could cause tissue damage on contact.

Requirements:

Conduct Exposure Assessments
Personal Protective Equipment
Work with Cryogens

Hazard Notes:

Standard cryogenic liquid dewars, cryostats and superconducting magnets may be used as sample environment equipment.

Control Notes:

1) Appropriate cryogenic training will be required for staff and users who will use cryogens. Complete the SNS Cryogenic and ODH Areas Training, Module (00051689).

2) Magnet use is subject to ACGIH boundary controls and SBMS rules

3) Appropriate PPE will be used.

Locations:

All Locations

Attachments:

None.

12.4

This operation involves elevated thermal hazards that could cause tissue damage on contact or a laboratory heating operation.

Requirements:

Fire Protection, Prevention & Control
Personal Protective Equipment

Hazard Notes:

EQ-SANS uses samples equipment that can heat up to high temperatures. For examples, as high as 130C for the peltier block systems and 300C for the furnace system. Touching the furnace or trying to remove samples from the furnace while it's at high temperature (T>40C) may cause tissue damage on contact

Control Notes:

All potential thermal hazards are contained within the instrument enclosure.

Only personnel specifically trained to use the equipment may perform sample manipulations.

Before removing the samples from the sample environment that operates at elevated temperatures, wait until the sample cools down to 40C or lower.

Wear appropriate PPE:

- Eyes – safety glasses at all times.
- Face – Add a full face shield where pressure releases or splash are anticipated.
- Hands – Loose fitting gloves that can be easily removed, not fitted with large cuffs in which liquids can collect (leather or cryogenic specialty fabric).
- Clothing – Long trousers (without cuffs preferred) to be worn outside shoes. The less porous (tight weaves) the fabric the better.
- Shoes – No open toed shoes.

Locations:

All Locations

Attachments:

None.

13.2

This operation involves confined spaces.

Requirements:

Confined Space

Hazard Notes:

The following areas meet all three criteria to be considered a confined space.

- Detector Tank
- Sample Area (Top)
- Sample Area (Bottom)

There are no potential physical or atmospheric hazards present inside of the confined spaces, making these areas non-permit required confined spaces.

The atmospheres of the spaces have been tested and is within safe operating limits. (see attached reclassification forms)

Control Notes:

- The detectors move back and forth in the tank. Movement is slow and access to the tank is controlled by the person working with the detector. When the detector is moving, line-of-sight is always maintained by the operator to ensure no one has entered the area.
- Ensure lighting, such as flashlights or headlamps, are available for use when entering these areas.
- A supply line from a nitrogen gas bottle feeds into the sample area. Ensure the valve on the bottle is closed during entry. Maintain control of the bottle during entry.
- If new gases are introduced to the space, contact a QHSP to assist with air monitoring, confined space classification, and to select appropriate controls before entering the space.

Locations:

All Locations

Attachments:

Detector Tank BL6 CONFINED SPACE PRE-ENTRY Determination-Reclassification form.docx
Sample Area (bottom) BL6 CONFINED SPACE PRE-ENTRY Determination-Reclassification form.docx
Sample Area (top) BL6 CONFINED SPACE PRE-ENTRY Determination-Reclassification form.docx

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13.3

This operation involves ergonomic hazards (e.g., lifting, pulling/pushing, vibration, posture, repetitive motion, excessive force, awkward positions, long duration, etc.).

Requirements:

Apply Ergonomics Controls
Conduct Exposure Assessments

Hazard Notes:

Sample environment and experimental set-ups may involve ergonomic hazards.

Frequent travel beneath the beam extender in the sample area that happens during sample environment changes may pose ergonomic hazards.

Work at the instrument console may pose ergonomic hazards.

Control Notes:

- As more workstations are created, we will evaluate work stations and follow guidelines
- Management encourages employee involvement and feedback in identification of existing and potential conditions that cause this disorder and encourages employees to report the conditions as early as possible. Workstations or task assessments are reviewed by an SOSD representative as soon as possible after they are reported.
- Engineering controls include: redesigning the workstation, and tools to reduce the demands of jobs involving excessive exertion, repetition, and awkward postures. Administrative controls include: establishing work/rest schedules, or training employees to use appropriate work methods.

Locations:

All Locations

Attachments:

None.

13.7

This operation involves the use of ladders or elevated work areas or platforms with an unprotected edge.

Requirements:

Fall Protection
Ladders
Scaffolding and Aerial Lifts

Hazard Notes:

The upper mezzanine is at same height as the building one (16'). The sample area mezzanine is halfway between the building mezzanine and the instrument floor.

The phone booth enclosure of the instrument at the sample area mezzanine has an SNS-standard sample environment mounting flange that is sufficiently large that a person could fall through it. The flange is normally covered by a large, stainless steel plug that itself has a smaller, secondary plug covering a smaller opening. The large plug, when the smaller secondary plug is in place, fully covers the opening of the flange. The opening covered by the secondary plug is not sufficiently large for a person to fall through.

Control Notes:

1) Methods to limit exposure to heights have been implemented, barriers and signs indicating the hazards will be in place.

2) Both the upper and lower mezzanine are enclosed by an approved guardrail system on all sides.

3) If hoisting activities require that a section of fencing be removed, the need for additional fall protection will be evaluated and authorized in the lift plan.

4) The aluminum cover plate will be placed on the flange if the access to the full opening is not required when working in the flange area.

Locations:

All Locations

Attachments:

None.

14.2

This operation involves work performed outside normal working hours (6am to 7pm, Monday-Friday).

Hazard Notes:

The instrument will be operated 24 hours a day during normal SNS beam operations.

Control Notes:

NSD:

1) During normal operation of the SNS/HFIR, the Instrument Hall Coordinator will be present 24 hours/day. Notification of the Instrument Hall Coordinator (865-241-4432) is required if working outside of normal working hours (7 pm to 6 am weekdays, holidays, and weekends). Name of the participant and brief summary of task to be provided to the IHCs. If the Instrument Hall coordinator is unavailable, contact the facility's control room (SNS Central Control Room (CCR) (865-576-1503) or HFIR Control Room (865-574-7035).

2. Work with hazardous chemicals (SBMS Chemical Safety Subject Area 1) will only be performed during normal working hours or with another staff member present.

3. Hoisting & Rigging will only be performed during normal working hours or with another trained staff member present.

4. Work with class 3b or greater lasers will only be performed during normal working hours or with another staff member present.

5. No Bio Safety Level 2 work outside normal working hours unless prior approval by IBC work plan.

The NSD: Planning high-risk/high-consequence work after normal working hours can be found here.

Locations:

All Locations

Attachments:

None.

14.3

This operation requires a special backup system, critical engineering controls or monitoring equipment, or alarms or emergency response for addressing off-normal conditions (e.g. emergency power, a failure detection system, or redundant equipment).

Requirements:

Engineering Design

Hazard Notes:

Some systems require battery backup.

The IPPS and BEAMLINE computer systems must have an alternate power source to allow appropriate response operation time to shut down the instrument safely in case of a catastrophic power failure.

Multiple alarms exist for different systems, including radiation alarms, oxygen-deficiency alarms, sample environment equipment (temperature, pressure, etc.), vacuum systems (pressure), detector high voltage, and data collection (time outs or other programing failures associated with data scans).

Control Notes:

UPS battery backup systems are utilized for instrument computer and control systems, and some sample environment equipment to bring them into "safe mode" in case of power interruption.

1.) Universal power supplies are installed with limited back-up power to allow for safe closure of the primary and secondary shutters and safe stoppage of all instrument components in motion.

2.) Crucial sample environment systems that need alternate power in case of a system power failure should incorporate system specific back-up power supplies.

Shielding and other credited controls are not removed or changed without approval from the SNS RSO.

Instrument Hall Coordinators monitor alarms continuously when SNS is operating and follow approved response procedures.

Locations:

All Locations

Attachments:

None.

15.1

This operation involves packaging or transporting of DOT hazardous materials (chemicals, biological materials, samples/specimens, RCRA waste, etc.) or DOT radioactive materials that have an off-site destination or originate off-site.

Requirements:

Commercial Motor Vehicle
Export Control Compliance
Off-Site Transportation
Package/Transport Nonhazardous, Hazardous, and Radioactive Materials Off-Site

Hazard Notes:

Small quantities of hazardous and/or radiological materials may be transported/shipped to or from this facility, and to and from nearby on-site laboratories as part of sample preparation. Radioactive materials are not permitted in all labs, contact RCT and LSM for guidance.

Control Notes:

Regulated shipments must be cleared through Logistical Services, Transportation and Packaging Management Organization as required by material type and hazard.

Materials of Trade training can help participants identify regulated materials. Personal vehicles may not be used to transport chemicals, hazardous materials, or radiological materials off-site.

TMO must determine if an item is regulated as Materials of Trade training does not give permission to make determination.

All shipments and/or transportation of Hazardous Materials or pre- & post- irradiated samples, and stock chemicals must go through the ES&H/OPS group. ES&H/OPS group will contact TMO for determination of MOT status.

All shipping and releas activities of samples and equipment potentially involve export control, discussed in https://sbms.ornl.gov/sbms/SBMSearch/SubjArea/export/sa.cfm

Locations:

All Locations

Attachments:

None.

15.2

This operation involves packaging or transporting of DOT hazardous materials (chemicals, samples/specimens, RCRA waste, etc.) or DOT radioactive materials on-site only.

Requirements:

Commercial Motor Vehicle
On-Site Transportation of Hazardous Material
Transport Chemicals

Hazard Notes:

Very small quantities of hazardous and/or radiological materials may be transported between this facility and other ORNL labs.

Small research samples that may be hazardous may also be transported between buildings at ORNL.

All shipping and release activities of samples and equipment potentially involve export contol, discussed in https://sbms.ornl.gov/sbms/SBMSearch/SubjArea/export/sa.cfm

Control Notes:

Regulated shipments must be cleared through Logistical Services, Transportation and Packaging Management Organization as required by material type and hazard.

Materials of Trade training can help participants identify regulated materials. Personal vehicles may not be used to transport chemicals, hazardous materials, or radiological materials on-site.

TMO must determine if an item is regulated as Materials of Trade training does not give permission to make determination.

All shipments and/or transportation of Hazardous Materials or pre- & post- irradiated samples, and stock chemicals must go through the ES&H/OPS group. ES&H/OPS group will contact TMO for determination of MOT status.

Locations:

All Locations

Attachments:

None.

17.1

This work requires the project staff to prepare or modify engineering calculations, drawings or specifications that are to be traceable and on record. Engineering calculations, drawings or specifications must be traceable and on record if used for construction, fabrication, modification, installation, or acquisition of engineered systems, structures, or components and meet any of the following criteria:

- Have the potential to adversely affect the health and safety of staff, workers, the public, or the environment
- Failure could unacceptably impact program objectives
- Are currently recorded in the ORNL Engineering Files or Records
- Are or will be installed as an integral operating system in a new or existing UT-Battelle controlled facility

Requirements:

Engineering Design

Hazard Notes:

Modifications to certain instrument systems (shielding, PPS, ventillation, structure, and etc.) shall performed in accordance with approved drawings, documents and specifications.

Control Notes:

The appropriate SME or engineering will be consulted prior to making any changes that can impact engineered systems, structures or other key components. Drawings of all modifications will be maintained.

Locations:

All Locations

Attachments:

None.

19.0

This work involves planned modifications/alterations of experimental equipment or planned maintenance of equipment.

Requirements:

Configuration Management
Electrical Safety
Select, Use and Modify Laboratory Equipment

Hazard Notes:

Some equipment at the beamline will require periodic maintenance, primarily sample environment equipment.

Control Notes:

Engage Subject Matter Experts to plan/perform work that is not within RSS participants skill set/qualifications.

Locations:

All Locations

Attachments:

None.